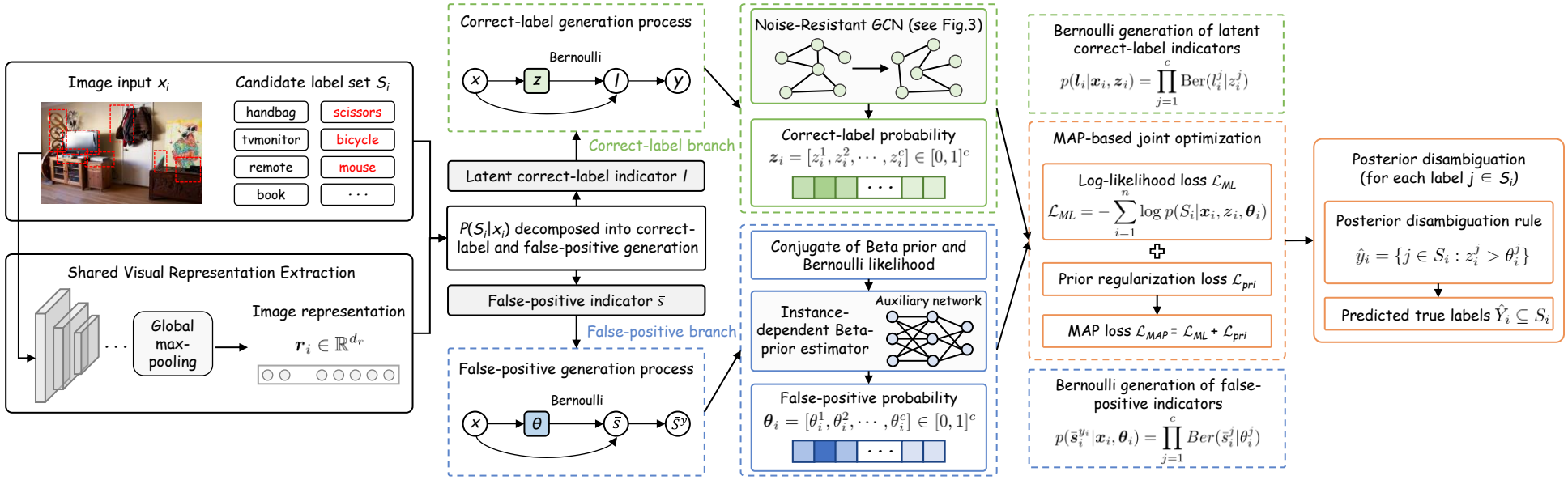

Image input x_i
Candidate label set S_i
handbag
scissors
tvmonitor
bicycle
remote
mouse
book
· · ·
Shared Visual Representation Extraction
Global max-pooling
Image representation
$\boldsymbol{r}_i \in \mathbb{R}^{d_r}$
Correct-label generation process
Bernoulli
x
z
l
y
Correct-label branch
Latent correct-label indicator l
$P(S_i|x_i)$ decomposed into correct-label and false-positive generation
False-positive indicator $\bar{s}$
False-positive branch
False-positive generation process
Bernoulli
x
θ
$\bar{s}$
$\bar{s}^y$
Noise-Resistant GCN (see Fig.3)
Correct-label probability
$\boldsymbol{z}_i = [z_i^1, z_i^2, \cdots, z_i^c] \in [0,1]^c$
Conjugate of Beta prior and Bernoulli likelihood
Instance-dependent Beta-prior estimator
Auxiliary network
False-positive probability
$\boldsymbol{\theta}_i = [\theta_i^1, \theta_i^2, \cdots, \theta_i^c] \in [0,1]^c$
Bernoulli generation of latent correct-label indicators
$p(\boldsymbol{l}_i|\boldsymbol{x}_i, \boldsymbol{z}_i) = \prod_{j=1}^{c} \mathrm{Ber}(l_i^j|z_i^j)$
MAP-based joint optimization
Log-likelihood loss $\mathcal{L}_{ML}$
$\mathcal{L}_{ML} = -\sum_{i=1}^{n} \log p(S_i|\boldsymbol{x}_i, \boldsymbol{z}_i, \boldsymbol{\theta}_i)$
Prior regularization loss $\mathcal{L}_{pri}$
MAP loss $\mathcal{L}_{MAP} = \mathcal{L}_{ML} + \mathcal{L}_{pri}$
Bernoulli generation of false-positive indicators
$p(\bar{\boldsymbol{s}}_i^{y_i}|\boldsymbol{x}_i, \boldsymbol{\theta}_i) = \prod_{j=1}^{c} Ber(\bar{s}_i^j|\theta_i^j)$
Posterior disambiguation (for each label $j \in S_i$)
Posterior disambiguation rule
$\hat{y}_i = \{j \in S_i : z_i^j > \theta_i^j\}$
Predicted true labels $\hat{Y}_i \subseteq S_i$